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Synthesis of 3-Keto Pyridines from the Conjugated Allenone – Alkynylamine Oxidative Cyclization Catalyzed by Supported Au Nanoparticles

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Dedicated to professor J. K. Gallos on the occasion of his retirement.

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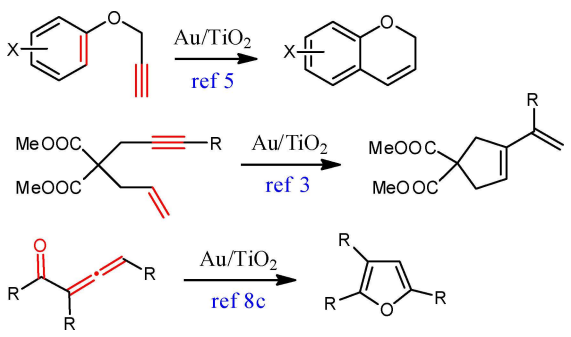
Abstract: Recyclable supported Au nanoparticles on TiO_2 catalyze the cyclization of *N*-propargyl or *N*-homopropargyl β-enaminones followed by dehydrogenation (aromatization) leading to substituted 3-keto pyridines or 4-picolines in very good yields. This pathway is in contrast to their known cyclization in the presence of Au(I) or Au(III) catalysts which provides 1,4-oxazepines, instead. The enaminones are formed in situ upon mixing a conjugated allenone or allenyl ester with the alkynylamine, thus the pyridine-forming transformation is typically a one pot process.

Keywords: Au nanoparticles; heterogeneous catalysis; cyclization; *N*-alkynyl β-enaminones; pyridines

Supported Au(0) nanoparticles (Au NPs) have been proven as excellent catalysts in cycloisomerization transformations, where the activation of *sp*-carbon π bonds (alkynes or allenes) on the low coordinated electrophilic Au atoms at the corners or edges of nanoparticles[1] triggers intramolecular nucleophilic attack by carbon nucleophiles (Scheme 1).[2] Thus, simple 1,6-enynes,[3] *N*-((1*H*-indol-3-yl)methyl)propiolamides (*N*-1,6-enynes),[4] aryl propargyl ethers,[5] or ω-alkynylfurans[6] undergo cyclization in the presence of catalytic amounts of supported Au nanoparticles. In addition, the activation of an alkyne or allene by Au NPs may trigger inter or intramolecular nucleophilic attack by nitrogen,[7] oxygen,[8] or other[9] nucleophiles.

These modes of reactivity were rather unanticipated, given that the same transformations are also smoothly catalyzed by cationic variants of metals as activating π acids, e.g. Au(I) or Au(III).[10] However, it has been shown theoretically that a neutral Au(0) cluster can strongly interact to an alkyne, and the triple bond is activated toward nucleophilic attack, with the magnitude of π back-donation in the triple bond – Au interaction to be more significant than σ donation.[11]

Our latest report, regarding the smooth Au/TiO_2-catalyzed cycloisomerization of conjugated allenones into furans,[8c] urged us to examine the possible Au-catalyzed cyclization of *N*-propargylic β-enaminones (*N*-1,5-enynes),[12] compounds that could be formally obtained from an uncatalyzed reaction between allenones and propargylamine.[13] Indeed, propargylamine adds quantitatively to the model allenone **1** in dioxane as solvent within 10 min at 25 °C, yielding exclusively the (Z)-enaminone **1′**. The observed Z-geometry is in accordance to similar enaminones in the literature.[14] The possible Au NP-catalyzed cyclization of *N*-1,5-



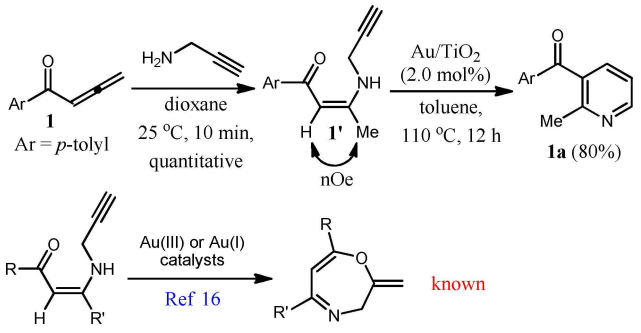
Scheme 1. Selected examples of cycloisomerization reactions catalyzed by Au/TiO_2 from our group.

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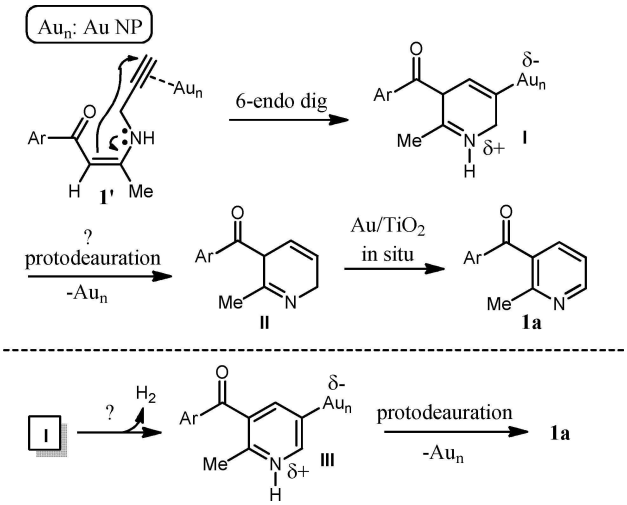
enyne **1′** was then examined (Scheme 2). While no reaction was seen under refluxing conditions in low boiling point solvents (e.g., DCM, THF), to our delight, in the presence of Au/TiO_2 (2 mol%) and refluxing toluene as solvent (110 °C), **1′** was completely converted into 3-keto pyridine **1a** after 12 h, which was isolated after column chromatography in 80% isolated yield.[15] In various other solvents, and at lower temperatures around 70–80 °C (including toluene) the cyclization proceeds sluggishly (up to 15% after 20 h). No 5-membered ring product (pyrrole) was formed from a possible 5-exo cyclization mode. Notably, the catalytic cyclization of *N*-propargyl β-enaminones in the presence of Au(I) or Au(III) substances leads to 1,4-oxazepines, via a 7-exo-dig pathway[16] (Scheme 2), or to 5-membered ring 3-methylene-1-pyrrolines in the presence of arynes and a Au(I) catalyst.[17] The structurally similar *N*-tosyl-*N*-propargyl enaminones lead to pyrroles using homogeneous Au(I) catalysts.[18] In this case it was suggested that cyclization is preceded by an amino-Claisen rearrangement forming an intermediate α-allenyl β-enaminone. As the uncatalyzed reaction of formation of the *N*-propargyl enaminone **1′** is very fast, we were able to achieve the synthesis of keto pyridine **1a** in a simple one pot operation. Thus, premixing 1 equiv. of allenone with 1.1 equiv. of propargylamine in toluene at 25 °C for 10 min, and then adding the catalyst followed by heating, resulted to the same outcome, as in the case of using directly **1′**. In fact, premixing the reactants for 10 min is more than enough, otherwise unreacted allenone **1** should had been instantaneously isomerized by Au/TiO_2 into 2-(*p*-tolyl)furan.[8c] The catalytic system (Au/TiO_2) is recyclable by simple filtration, washing and drying at 90 °C for 2 h. It was reused in five consecutive runs with little loss of activity. Thus, in the 5th run, the reaction took 15 h to go to completion, which is attributed to the partial loss of catalyst in each recycling procedure. As a further evidence of the heterogeneous nature of our transformation, we found that the cyclization of *N*-propargyl enaminones does not proceed at all in the hot filtrate of another cyclization reaction, which implies that Au leaching into the solution is essentially minimal, if any.

The transformation **1′**→**1a** indicates that Au NPs activate the alkyne that then is susceptible to a 6-endo nucleophilic attack by the enaminone double bond leading to intermediate **I** (Scheme 3). This intermediate could undergo protodeauration, which releases the dihydropyridine **II** from the chemisorbed species **I** and regenerates the catalyst. We could not, however, detect **II** by GC-MS at any stage of the reaction, which possibly implies that the dehydrogenation of the dihydropyridine (or any isomer of it) into pyridine **1a** may occur faster than the initial cyclization. Alternatively and more likely, chemisorbed **I** could undergo dehydrogenation into aromatic **III**, which after protodeauration should lead directly into **1a**. As the reaction, also takes place with the same rate under inert conditions, we conclude that the dehydrogenation of intermediate **I** (Scheme 3) is an acceptorless H_2 elimination process. Preceding relevant examples of Au NP-catalyzed dehydrogenations are well known in the case of 1,2-dihydroquinolines, that are oxidized into quinolines in the presence of Au/TiO_2.[19] In intermediate **I** we consider partial and not full charges, based on previous calculations on the bonding nature of chemisorbed intermediates on Au clusters.[20]

The cyclization of *N*-propargyl enaminones by other catalytic systems is known so far only under homogeneous catalysis conditions and can lead either to 5- or 6- membered ring products. In the presence of 20–40 mol% CuBr, pyridines are formed in moderate to good yields, yet under catalysis by a base, pyrroles were seen, instead.[21] Cyclization/aromatization has been achieved using stoichiometric amounts of CuI, as



Scheme 2. The Au/TiO_2-catalyzed cyclization of a model compound **1′**, and the known[16] Au(I) or Au(III)-catalyzed cyclization of *N*-propargylic β-enaminones.



Scheme 3. Possible mechanisms in the Au-catalyzed formation of 3-keto pyridine **1a** from *N*-propargyl enaminone **1′**.

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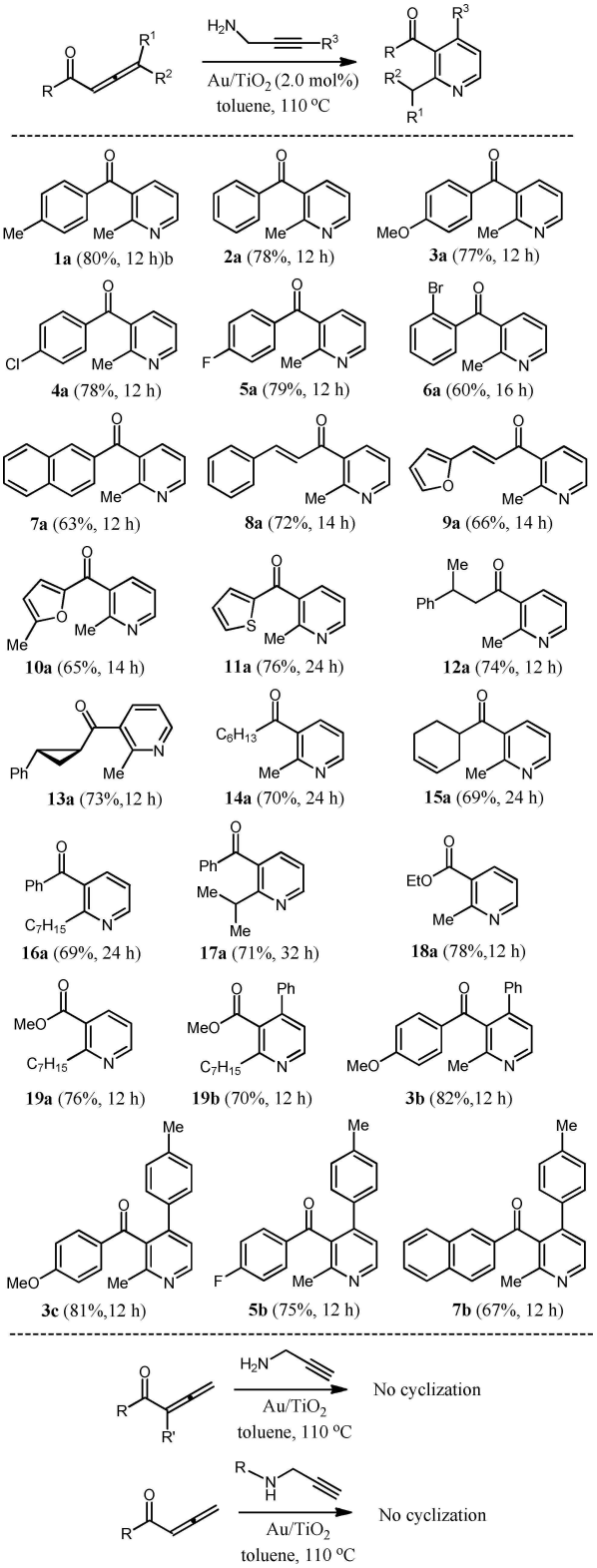
the oxidation of the cyclization products (dihydropyridines) into pyridines is not catalytic, with Cu(I) being reduced into Cu(0).[22] If the N atom of *N*-propargyl enaminones is trisubstituted, 1,6-dihydropyridines can be also formed in the presence of a Cu(I)-catalyst (3–10 mol%).[23] Similar results were obtained in the presence of $AgNO_3$ (10 mol%).[24] Using $CuCl_2$ (20 mol%) and a base, *N*-propargyl enaminones cyclize into pyrroles.[25] Finally, with stoichiometric amounts of $ZnCl_2$, the cyclization leads to the 7-membered ring 1,4-oxazepines.[26] Apart from the metal-catalyzed protocols, upon treatment with a base in the presence of a suitable nucleophile or electrophile, *N*-propargyl enaminones lead to substituted pyridines,[27] but not to 3-keto pyridines, as in our case.

Having optimized the reaction conditions (2 mol% Au/TiO_2, toluene as solvent, 110 °C), we examined the scope and limitations of this transformation. Thus mixing a series of conjugated allenyl ketones (1 equiv.) with 1.1 equiv. of propargylamine for 10 min and then adding the catalyst, followed by heating for a certain period of time, we found that the reaction is general. The isolated yields of pyridines are very acceptable and vary between 65–80% (Table 1). Using aryl substituted internal propargylamines, formation of pyridines also occurs readily under the same reaction conditions, despite the substantial increase of bulkiness on the triple bond. Note that, apart from conjugated allenones, allenyl esters work equally well (products **18a**, **19a** and **19b**).

Regarding allenones, the only non-productive substrates are those having a substituent at the germinal position with respect to the carbonyl group (see the bottom part of Table 1). Such substituents generate a quaternary center in the possible intermediate **I**, and thereby it cannot be aromatized into **III** (see the proposed mechanism in Scheme 3). Additionally, secondary propargyl amines (R = $-CH_2C{\equiv}CH$ or Me) do not lead to cyclized products (bottom part of Table 1), despite the fact that the corresponding *N*-propargyl enaminones are readily formed. In this case a hypothetical alkyldeauration would be an unfavorable process to occur.

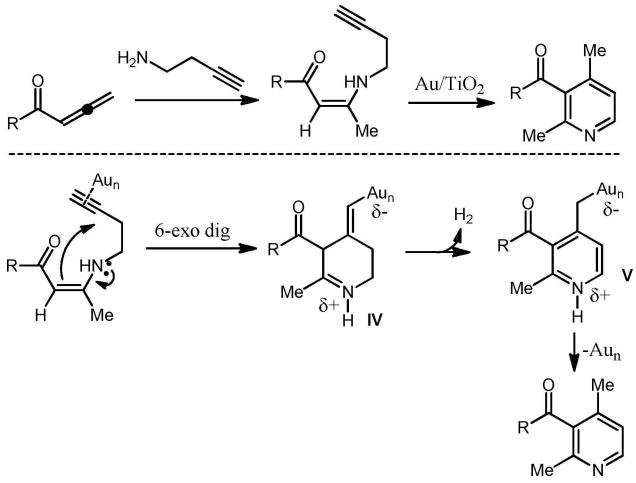
Following the very fruitful results of the synthesis of pyridines using propargylamines,[28] we then examined the parent homopropargylamine ($HC{\equiv}CCH_2CH_2NH_2$). In this case a 6-exo cyclization pathway in the corresponding *N*-homopropargyl enaminones could lead to 3-keto 4-picolines. Indeed, the acyclic enaminones are readily formed (see compound **1″** as a typical example in pages 43–44 of the Supporting Information), and are cyclized in the presence of 2 mol% Au/TiO_2, forming the 4-methylpyridines (4-picolines) in very good isolated yields. Our proposed mechanism appears in Scheme 4, in which most likely intermediate **IV** is dehydrogenated into **V** and after protodeauration the 3-keto 4-picoline

Table 1. Synthesis of pyridines from the cyclization/dehydrogenation of in situ generated *N*-propargyl enaminones catalyzed by Au/TiO_2.[a]



[a] Isolated yields after chromatography (0.2 mmol scale).
[b] 77% yield in 1.0 mmol scale.

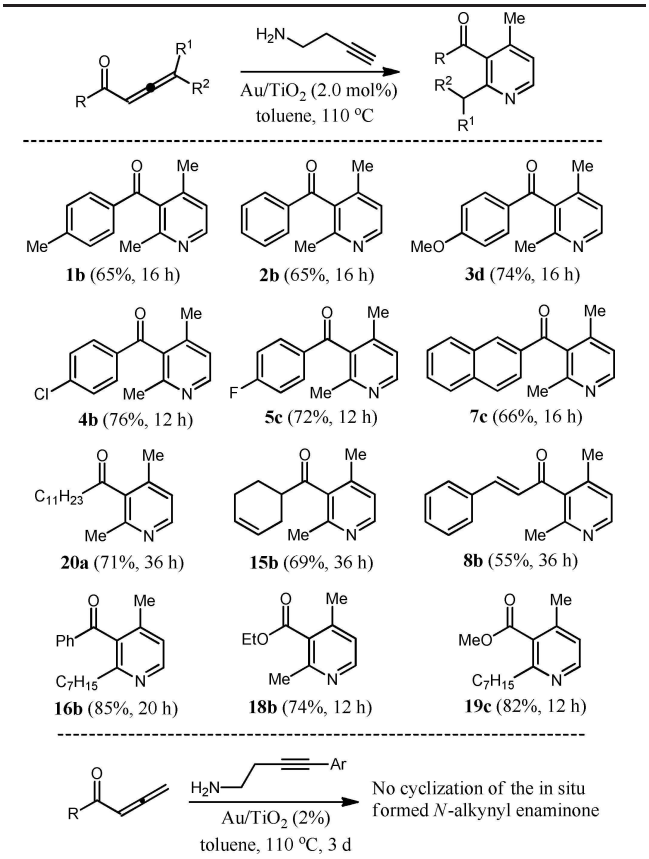
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Scheme 4. The Au-catalyzed oxidative cyclization of an allenone with homopropargylamine and a proposed mechanism.

is released. In Table 2 we provide a series of such examples. Unfortunately, *N*-alkynyl enaminones generated from simple allenones and aryl-substituted homopropargylamines, $ArC{\equiv}CCH_2CH_2NH_2$, (see compound **1**‴ as a typical example in the last page of the Supporting information), are completely unreactive, even if applying forcing conditions, which implies that the 6-exo cyclization mode is sensitive to steric factors, in contrast to the corresponding 6-endo process shown in Table 1.

Table 2. Synthesis of 4-picolines from the cyclization/dehydrogenation of in situ generated *N*-homopropargyl enaminones catalyzed by Au/TiO_2.[a]



[a] Isolated yields after chromatography (0.2 mmol scale).

In conclusion, we show herein the catalytic ability of supported Au nanoparticles on TiO_2, in catalyzing the oxidative cyclization in the reaction between conjugated allenones or allenyl esters and propargylamines or homopropargylamine, forming substituted 3-keto pyridines and 4-picolines. The observed cyclization pathway is different from that occurring under homogeneous ionic Au-catalysis. Our catalytic system is readily available, and the process is purely heterogeneous, allowing its recycling and reuse.

Experimental Section

Allenones **1**–**13**, **15**, **17** and **20** were available from previous studies in our lab.[8c] Allenyl ester **18** is commercially available, while compounds **16** and **19** were prepared following a known procedure.[29] Apart from the commercially available propargylamine and homopropargylamine, their aryl-substituted counterparts were synthesized as described in the Supporting Information, using known literature procedures.[21,30]

General procedure of the Au/TiO_2-catalyzed cyclization: To a vial containing a solution of 0.2 mmol of allenone or allenyl ester in toluene (0.5 mL) are added 0.22 mmol of the alkynylamine. After stiring for 10 min at 25 °C, Au/TiO_2 (74 mg, 2.0 mol% in Au) is added and the mixture is heated to 110 °C for a certain period of time until the reaction was complete (see Tables 1 and 2). The slurry is filtered with the aid of dichloromethane under a low pressure through a short pad of silica gel and the filtrate is evaporated to afford the pyridines. The cyclization was also performed on a larger scale using 158 mg (1.0 mmol) of allenone **1**, 71 μL of propargylamine (1.1 mmol), toluene (3 mL) and Au/TiO_2 (370 mg, ~3.7 mg of Au, as the catalyst contains 1 w/w% Au). After 12 h at 110 °C the reaction was complete and following the above workup, pyridine **1a** was isolated in 78% yield (162 mg) after column chromatography, using petroleum ether/ethyl acetate = 2/1 as eluent.

Acknowledgements

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